

ls, date, cal commands

MODULE 1: GETTING STARTED & NAVIGATING THE SYSTEM

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Introduction

- The ls, date, and cal commands are essential Linux/Unix utilities for viewing directory contents, managing system time, and displaying calendars.
- The **ls** command lists files and directories in the current or specified path.
- The **date** command shows or configures the system's current date, time, and timezone.
- The **cal** command outputs a text-based calendar directly in the terminal.

ls command

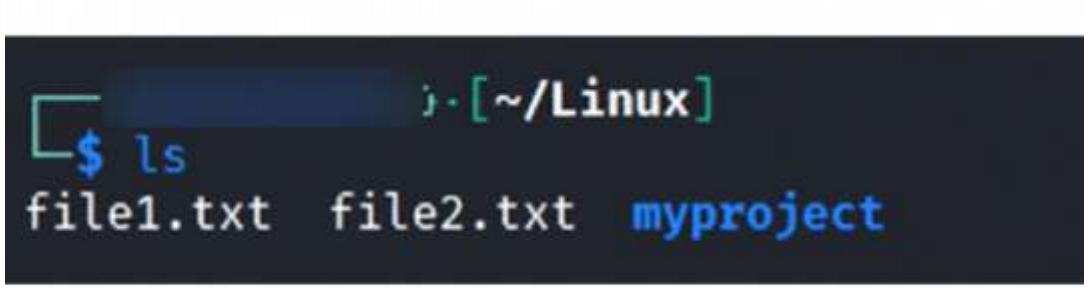
The ls command is used to list all files and directories in the Linux terminal.

- Displays the contents of the current working directory or a specified path.
- Lists items in alphabetical order by default.
- The command can be customized with flags to display file permissions, ownership, size, and modification date.
- Helps users view file and directory names with optional details.

Examples of Listing Files and Directories

- Type the basic ls command in the terminal to display all files and directories in the current working directory.

Command: ls



Various options of ls command

Given below are some commonly used options with the ls command to display different types of files and directories.

Syntax: ls [option] [file/directory]

- Open Last Edited File Using `ls –t`
- Display One File Per Line Using `ls -1`
- Display All Information About Files/Directories Using `ls –l`
- Display File Size in Human Readable Format Using `ls –lh`
- Display Directory Information Using `ls –ld`
- Order Files Based on Last Modified Time Using `ls –lt`

Various options of ls command

- Order Files Based on Last Modified Time (In Reverse Order) Using `ls –ltr`
- Display Hidden Files Using ls –a (or) ls –A
- Display Files Recursively Using ls – R $ ls -R /etc/apt
- Display File I-node Number Using `ls –i`
- Hide Control Characters Using `ls –q`
- Display File UID and GID Using `ls –n`
- Visual Classification of Files with Special Characters Using `ls –F`
- Visual Classification of Files with Colors Using `ls –F`
- Change the way time information is displayed using –time-style flag

Display File Size in Human Readable Format Using `ls –lh`

-rw-rw-r-- 1 user user 1176 Feb 16 00:19 test

First character hypen(-) specifies the type of the file.

File Type Indicators

- (normal file): Represents a regular file.
d (directory): Indicates that the file is a directory.
s (socket file): Indicates that the file is a socket.
l (link file): Indicates that the file is a symbolic link.

If you want see the file size, permissions, owner and date than use the below command: **ls –lh**

-l: lists files in long format (with permissions, owner, size, and date)
-h: shows the file size in a human-readable format like KB, MB, or GB instead of just raw bytes

File listing field explanations

[Field 1]: File Permissions

The next characters specify the file permissions. Every three characters represent read, write, and execute permissions for the user (owner), group, and others respectively.

Example: -rw-rw-r-- means read-write for user, read-write for group, and read-only for others.
Example: -rwxrwxrwx means full access for all.

[Field 2]: Number of Links

Specifies the number of hard links associated with the file.
Example: 1 indicates there is only one link to the file.

[Field 3]: Owner
Shows the user who owns the file.
Example: the owner could be a username like user.

[Field 4]: Group
Indicates the group to which the file belongs.
Example: user group.

[Field 5]: File Size
Displays the size of the file in bytes.
Example: 1176 means the file size is 1176 bytes.

[Field 6]: Last Modified Date and Time
Shows when the file was last modified.
Example: Feb 16 00:19 indicates the last modification time.

[Field 7]: File Name
Displays the name of the file.
Example: the file name could be 1.c.

Display Directory Information Using `ls –ld`

When you use **“ls -l”** you will get the details of directories content. But if you want the details of the directory then you can use -d option

For example, if you use ls -l /etc will display all the files under the etc directory.
But, if you want to display the information about the /etc/ directory, use -ld option as shown below.

ls –l /etc [-l] means long listing format (it shows permissions, owner, size, etc).

If you want to see the details about a directory itself (and not the files inside it),
you can use:

ls –ld /etc [-d] tells the system to show info about the directory itself, not what’s inside it.

date command

The date command in Linux allows the user to display the current date and time in a variety of formats and set the system date and time.

Uses of the date Command are :-

- Displaying the Current Date and Time
- Setting the System Date and Time
- Customizing the Output Format
- Performing Date Calculations
- Displaying Time in Different Time Zones

Displaying the Current Date and Time

The date command displays the current date and time, including the abbreviated day name, abbreviated month name, day of the month, the time separated by colons, the time zone name, and the year.

Syntax: date

```
yoges@LAPTOP-DCISH7AJ:/mnt/c/Users/yoges$ date
Wed Oct 28 01:00:12 UTC 2026
yoges@LAPTOP-DCISH7AJ:/mnt/c/Users/yoges$
```

Here, Unix system is configured in Pacific daylight time and displaying the day, month, date, time, timezone, and year.

Various options of date command

Given below are some different ways to view date and time using the date command options.

Syntax

```
date [OPTION]
date [+FORMAT]
date [-u|–utc|–universal] [MMDDhhmm[[CC]YY][.ss]]
```

OPTION: Refers to different flags & arguments that modify the behavior of the command.
FORMAT: Refers to "how you want to display" the date, using format specifiers. (such as %Y, %m, %d).
u , utc, and universal: Tells date to use UTC (Coordinated Universal Time) instead of local time.
mm: Refers to Month (01-12)
DD: Refers Day (01-31)
hh: Refers Hour (00-23)
MM: Refers Minutes (00-59)
CC: Refers Century (19, 20, 21, etc.)
YY: Refers Year (00-99)
ss: Seconds

Display Time

Syntax: date –u

The **-u** option with the date command allows you to display the current time in GMT (Greenwich Mean Time) or UTC (Coordinated Universal Time) instead of your system's local time zone.

```
yoges@LAPTOP-DCISH7AJ:/mnt/c/Users/yoges$ date -u
Wed Oct 28 01:10:43 UTC 2026
yoges@LAPTOP-DCISH7AJ:/mnt/c/Users/yoges$
```

Syntax: date –date=" datestring " + FORMAT

Displays the given date string in the format of date. But this will not affect the system's actual date and time value. Rather it uses the date and time given in the form of string.

```
yoges@LAPTOP-DCISH7AJ:/mnt/c/Users/yoges$ date -d "10/28/2026"
Wed Oct 28 00:00:00 UTC 2026
yoges@LAPTOP-DCISH7AJ:/mnt/c/Users/yoges$ date -d "Oct 28 2026"
Wed Oct 28 00:00:00 UTC 2026
yoges@LAPTOP-DCISH7AJ:/mnt/c/Users/yoges$
```

Display Past Dates

Syntax: date –-date “2 year ago” - Displaying the date and time of 2 years ago

```
yoges@LAPTOP-DCISH7AJ:/mnt/c/Users/yoges$ date --date="2 year ago"
Mon Oct 28 01:17:01 UTC 2024
yoges@LAPTOP-DCISH7AJ:/mnt/c/Users/yoges$
```

Syntax: date –-date “5 sec ago” - Displaying the date and time of 5 seconds ago

```
yoges@LAPTOP-DCISH7AJ:~/projects/qa-mock-test$ date --date "5 sec ago"
Wed Oct 28 11:28:58 UTC 2026
```

Syntax: date –-date “yesterday” - Displaying the date and time of previous day.

Syntax: date –d “next tuesday” - Displaying the date and time of upcoming particular week day.

Syntax: date –d “tomorrow” - Displaying the date and time next day.

Syntax: date –d “1 year” - Displaying the date and time of upcoming particular week day.

List of Format Specifiers used with Date command

%D: Display date as mm/dd/yy.
%d: Display the day of the month (01 to 31).
%a: Displays the abbreviated name for weekdays (Sun to Sat).
%A: Displays full weekdays (Sunday to Saturday).
%h: Displays abbreviated month name (Jan to Dec).
%b: Displays abbreviated month name (Jan to Dec).
%B: Displays full month name(January to December).
%m: Displays the month of year (01 to 12).
%y: Displays last two digits of the year(00 to 99).
%Y: Display four-digit year.
%T: Display the time in 24 hour format as HH:MM:SS.
%H: Display the hour.
%M: Display the minute.
%S: Display the seconds.

Syntax and Examples

Syntax: date +%[format-option]

Examples:

- **date "+%D"** – Display Date in MM/DD/YYYY Format
- **date "+%D %T"** – Display Date and Time in MM/DD/YY HH:MM:SS Format
- **date "+%Y-%m-%d"** – Display Date in YYYY-MM-DD Format
- **date "+%Y/%m/%d"** – Display Date in YYYY/MM/DD Format
- **date "+%A %B %d %T %y"** - Display Full Date and Time with Day, Month, Date, Time and Year

cal command

- 'cal': Shows current month calendar on the terminal.
- ‘ncal': Shows current month calendar on the terminal with the current date highlighted.
- 'cal -y': Shows the calendar of the complete current year with the current date highlighted.
- 'cal [month] [year]': Shows calendar of selected month and year. For example, 'cal 08 2026' displays the calendar for August 2026.
- 'cal [year]': Shows the whole calendar of the year. For example, 'cal 2026' displays the calendar for the year 2026.
- 'cal [year] | more': When year is not be visible in the same screen use 'more' with 'cal' use spacebar to scroll down.
- 'cal -3': Shows calendar of previous, current and next month
- 'cal -j': Shows the calendar of the current month in the Julian calendar format not in the default Gregorian calendar format.
- NOTE: In Julian calendar format, the date does not reset to 1 after every month's end i.e. after 31st Jan, Feb will start as 32nd Feb, not as 1st Feb. But in the Gregorian calendar format, the date is reset to 1 after every month's end i.e after 31st Jan, Feb will start as of 1st Feb.